

WILLIAM NZIVE

Computer Science & Engineering | Machine Learning, Embedded Systems & Robotics

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EDUCATION

University of California, Davis — B.S. Computer Science & Engineering **Expected Jun 2030**
School of Engineering and Sciences — High School Diploma, Salutatorian **2026**

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, Assembly, Java, C#, SQL, JavaScript

ML / AI: CUDA, model construction and training, fine-tuning, computer vision, signal processing, RAG, model evaluation

Systems: ESP32, ESP-IDF, Linux, firmware optimization, networking, Docker, Git, bootloaders, operating systems

Engineering: Creo, SolidWorks, Onshape, KiCad, CNC machining, 3D printing, MIG/TIG welding

RESEARCH INTERNSHIP

UC Santa Cruz Science Internship Program (SIP) — *Research Intern* **2025**

- Worked with graduate researchers during a nine-week internship focused on Wi-Fi channel-state information (CSI), sensing, and machine learning.
- Proposed an original direction using Wi-Fi signals for audio sensing/reconstruction; built deep-learning pipelines, configured experiments, collected data, and evaluated classification performance.
- Identified packet sampling as the primary experimental bottleneck, turning failed classification into the roadmap for continued data-acquisition and embedded-systems work.

INDEPENDENT RESEARCH & PROJECTS

Wi-Fi CSI Audio Sensing — *Independent Researcher* **2025–Present**

- Continued the SIP project independently, developing a low-cost ESP32-based system for extracting audio information from RF channel-state measurements.
- Reworked performance-critical firmware and the collection stack below normal Wi-Fi abstractions, increasing capture from about 300 to 7,000 packets/s.
- Built and iterated neural architectures and signal-processing pipelines; continue refining filtering, alignment, collection methods, and model design toward a publication-ready prototype.
- Work includes experimental design, firmware optimization, data acquisition, model training/evaluation, failure analysis, and technical documentation.

Parkinson's DaTscan Classification — *Machine Learning Challenge* **2026–Present**

- Built ML pipelines for 3D DaTscan classification, testing preprocessing, augmentation, learned image representations, routing, and model ensembles.
- Used fold-based validation and held-out testing to catch overfitting and compare changes before carrying them forward.
- Current standing: 23rd of 864 registered competitors (Top 2.7%).

Hybrid Language Model Research **2026–Present**

- Experimentation with efficient diffusion between language models of different architectures and sizes.

- Repurposed techniques including JEPA and DINO to aid in the efficient diffusion process.
- Compared language-model loss, KL divergence, intermediate representations, and model behavior across experiments.

Custom Operating System & Programming Language

2024

- Built a bootloader, operating-system prototype, and custom programming language while teaching myself lower-level computer architecture.
- Implemented a simple Snake game within that OS in Assembly and used what I learned later in firmware optimization.

Vulcan — Open-Source Robotic Arm

2025–2026

- Developed a low-cost, open-source large robotic arm, including CAD, component selection, bill of materials, and build documentation.
- Iterated the design around low-cost hardware and manufacturing methods intended to make the system practical to reproduce.

Low-Cost 3D-Printed Integrated Robotic Joints

2026–Present

- Developed low-cost integrated robotic joints built around 3D-printed components.
- Designed and iterated the joint, transmission, bearings, actuator packaging, and mechanical interfaces to increase torque capacity while keeping the design inexpensive to manufacture.

LEADERSHIP & ENGINEERING

FRC Team 3598 — SEStematic Eliminators — *Team Captain*

2025–2026

- Led a 40+ member robotics team through design, fabrication, programming, electrical integration, competition preparation, and outreach.
- Mentored students in Java, Git, CAD, machining, and 3D printing; helped the team earn the FIRST Impact Award and qualify for the FIRST World Championship.

SMUD Solar Regatta — *Programming Captain*

2025

- Reverse-engineered poorly documented controller libraries in C# and wrote custom steering and throttle logic for a solar-powered competition boat.
- Traced a sensor-interface failure across programming and electrical systems and used it to improve cross-team documentation.

HACKATHONS & EXHIBITIONS

Hack Club Outpost — 72-Hour Hardware Hackathon, San Francisco | **2026**

Open Sauce — Project Exhibitor, San Mateo | **2026**

Hack Club Fallout — Hardware Hackathon Participant | **2026**

Hack Club Daydream — Global Game Jam | **2025**

Hack Club Shiba Arcade — Game Development / Arcade Program | **2025**

REFERENCE

Dr. Miguel Lopez — High School Counselor & Robotics Mentor | miguel-lopez2@scusd.edu